PABLO BANO BENITO

C# / C++ Software Engineer | Systems, Tools & Maintainable Runtime Software

Spain | Remote only | EU work authorization | pablo.bano.benito@gmail.com | Portfolio | LinkedIn | English / Spanish

PROFILE

C# / C++ software engineer with 4+ years of C# engineering experience across production game development, tools, debugging and modular runtime systems. Strongest evidence: end-to-end feature ownership on a multiplayer, physics-driven Unity 6 production and a private custom C++ engine case study covering architecture, resources, editor tooling, scripting, audio and Linux packaging. Best fit for software engineering roles where maintainability, debugging, architecture and production ownership matter.

TECHNICAL SKILLS

Strong evidence: C#, C++, Unity 6, debugging, refactoring, runtime systems, modular architecture, code review, production support

Production workflow: Git, pull requests, Jira, TeamCity, QA validation, milestone stabilization, cross-functional delivery

Systems / tooling: Internal tools, component systems, resource management, profiling, memory / GC optimization, test scenes, feature validation

Engine / media stack: Custom C++ engine, OpenGL, Dear ImGui, Lua, Unreal Engine 4, FMOD, SoLoud, CMake / Linux packaging

AI-assisted workflow: ChatGPT, Copilot and Claude for codebase exploration, architecture review and edge-case checking; final code decisions remain human-reviewed

PROFESSIONAL EXPERIENCE

Programmer / Software Engineer | No Brakes Games - Human Fall Flat 2 | Mar 2022 - Oct 2025

Full-time C# production engineering on a multiplayer, physics-driven Unity 6 title with a 20-30 person multidisciplinary team. Work was organized around milestone delivery, feature ownership, pull requests, QA verification, TeamCity builds and production stabilization.

- Owned C# feature branches from ambiguous requirements through architecture, implementation, local validation, PR review, QA verification, TeamCity build checks and production merge.
- Designed modular, package-based Unity systems using asmdef assemblies, reusable components, interfaces / abstract classes, ScriptableObject-driven configuration and isolated feature test scenes.
- Built internal tools and runtime workflows for designers, audio and QA, including NDA-safe categories such as level-authoring utilities, validation scenes, timer/action scheduling, runtime debug support and level-scoped audio loading.
- Diagnosed multiplayer, physics and runtime behaviour issues by isolating repro cases, comparing host/client state, adding logs/assertions and validating edge cases before QA handoff.
- Improved runtime behaviour with Unity Profiler, Memory Profiler and ProfilerMarkers, reducing avoidable calls, managed allocations / GC pressure and heavy single-frame workloads where relevant.
- Fixed 100+ bugs and runtime issues across milestones, including gameplay defects, multiplayer issues, crash/freeze investigations, FPS/memory regressions and QA-reported production blockers.

SELECTED TECHNICAL PROJECTS

SufferEngine - Private custom C++ real-time engine | case study / video

- Co-developed a custom C++ engine with ECS-style Scene/GameObject/Component/System architecture, OpenGL rendering, DisplayList/Command render submission, ResourceManager, Lua scripting, SoLoud audio and Dear ImGui editor tooling.
- Worked across engine-level organization: runtime loop, transform hierarchy, resource lifetime, editor integration, third-party library integration, CMake/Linux support and AppImage-style packaging.
- Strongest portfolio evidence for C++ systems thinking; positioned as architecture/tooling work, not as a commercial engine or graphics-only specialization.

Oona the Druid's Path - Released Unreal Engine 4 / C++ team project | Steam

- Implemented player movement, AI behaviour trees, Blueprint-facing C++ APIs, combat, collectibles and level support in a multidisciplinary team project.
- Used Unreal Engine 4, C++, Blueprints and Perforce to deliver gameplay-facing systems for a released Steam title.

Magoon - Published Unity / C# Android roguelike | Google Play

- Built C# mobile game systems including grid movement, A* pathfinding, procedural dungeon structure, enemies, collectibles, reward loops and progression.

EDUCATION

BSc (Hons) Computer Science - Sheffield Hallam University - First Class Honours - 2021

Higher National Diploma (HND) in Computing - Video Games Programming - ESAT - 2017-2020